

Highlights

- Fishing communities rely heavily on fish stocks often threatened by overexploitation
- Fishing households make consumption and production decisions simultaneously
- We propose a dynamic model that includes fish as a common-pool resource
- The model is validated with data from a village in the Colombia Caribbean
- The calibrated model simulates policy effects on wellbeing and fishery sustainability
- Alternative economic activities can raise living standards without harming fish stocks

Understanding Livelihoods of Small-Scale Fishing Communities: A Dynamic Fishing Household Production Model

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Abstract

Small-scale fishing (SSF) communities in developing countries typically face high levels of poverty and vulnerability, coupled with a strong dependence on fishing resources. Fishing plays a central role as both a source of income and a means of ensuring food security. However, there is limited information on the relationships between household well-being and resource sustainability. This study contributes to a deeper understanding of SSF livelihoods by examining how key economic parameters influence the decision-making of fishing households. We develop a dynamic fishing household production model in which households make simultaneous decisions regarding consumption and production, incorporating the fact that fishing depends on a common-pool resource under open-access regime that fluctuates over time. The theoretical model is validated using data from the village of Barú, in the Colombian Caribbean. The calibrated model allows for simulations of various policy scenarios affecting SSF management and household well-being, both with and without considering changes in fish stock. The results highlight that improvement in alternative income opportunities, such as higher external wages, can simultaneously enhance household well-being and reduce fishing pressure on common-pool resources. Moreover, policies that integrate social protection mechanisms are essential if we are to prevent increases in poverty and food insecurity when conservation measures restrict fishing activities. Our model shows that policy evaluation must take into account the fact that households in developing countries make production and consumption decisions simultaneously. In the context of fisheries, this integrated decision-making can generate insights that can only be clearly articulated through a household production framework. Our findings offer insights for the design of policies that simultaneously support the sustainable use of marine resources (in biodiverse ecosystems) and foster the socioeconomic development of these communities.

Keywords: artisanal fisheries; sustainability; common-pool resources; intertemporal effects; developing countries.

JEL Codes: C63, C83, D13, J22, O13, Q22, Q56, Q57

1. INTRODUCTION

The sustainability of small-scale fisheries (SSF) has become an increasingly prominent issue in fisheries economics given the high level of biodiversity associated to the marine ecosystems where they operate and the vulnerability of fishing communities (Lancker et al., 2019; FAO & World Fish Center, 2008). SSF are responsible for approximately half of the annual global marine catch (FAO, 2024) and account for 90% of employment in the fisheries sector (FAO, 2024). Many of these jobs are concentrated in communities that depend heavily on fishing resources for their subsistence (Olale & Henson, 2013; Allison & Ellis, 2001).

In Colombia, a megadiverse country, up to 150,000 fishers and their households rely on fish resources, with one-third depending specifically on coastal fish stocks (OECD, 2016). These fishers typically lack access to formal employment, working independently and informally, with limited human and physical capital. While labor informality is widespread across Latin America, affecting more than half of the workforce, in the primary sector informality is even more pronounced, exceeding 75%, largely due to the absence of adequate labor markets.

Although marine fishing production, in terms of catch volume, has generally remained stable in temperate and outcropping areas and has shown a slight increase in tropical areas, the percentage of species populations exploited at biologically unsustainable levels rose from 10% in 1974 to 38% in 2021 (FAO, 2024). In the case of Colombia's marine fisheries, Rueda et al. (2019) report a decline in landed catch since 2006. This trend may be linked to reduced profitability in certain fisheries and low competitiveness (Merino et al., 2013), the degradation of the fish stocks—evident in a considerable number of marine species under some threat category (Ardila et al., 2002)—, and the increasing ecological footprint of fishing activities (Vargas-Morales et al., 2013).

The degradation of marine-coastal ecosystems, combined with stressors such as climate change and high levels of poverty in coastal areas, increases the vulnerability of artisanal fishing

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4 communities (Salas et al., 2011). SSF are typically associated with households with limited access
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6 to alternative sources of income, land, capital, and labor markets, often rendering them dependent
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8 on fish stocks—commonly managed under open-access regimes—that serve as both a source of
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10 income and a source of food. For these reasons, understanding the livelihoods of artisanal fishers
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12 and identifying key economic factors that affect fishing household decision-making is essential for
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14 designing policies that promote both the sustainable use of marine resources and the socioeconomic
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16 development of fishing communities.
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21 A key aspect of this understanding is recognizing that artisanal fishing must be analyzed not only
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23 based on the decisions of fishers as producers, but also on the household as an economic unit. Like
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25 rural households more broadly, fishing households participate simultaneously in goods and labor
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27 markets as producers and consumers. They allocate family labor across fishing and non-fishing
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29 economic activities, and household consumption needs, while also producing fish both for sale and
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31 for self-consumption. This dual role creates interdependencies between production and
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33 consumption decisions that differ fundamentally from predictions derived solely from consumer
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35 theory. For example, higher fish prices may not necessarily reduce fish consumption, as they also
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37 increase income from selling fish and thus consumption opportunities.
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43 By contrast, fishing models from traditional resource economics (e.g., Conrad and Rondeau, 2020)
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45 assume that fishers are firms maximizing profits from selling harvest without any consideration of
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47 self-consumption of fish as a source of well-being for the household, beyond income.
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51 Therefore, despite substantial interest in rural household interdependencies (Chang et al., 2012),
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53 empirical applications to fishing households remain scarce. Existing studies tend to analyze either
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55 production or consumption decisions, but not both jointly. Relevant exceptions include De la
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57 Montaña et al. (2015) and Maldonado et al. (2021), who examine integrated models for specific
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59 indigenous and coastal communities. The lack of integrated approaches limits our understanding
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4 of how artisanal fishing households respond to economic incentives and policy interventions,
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6 particularly in open-access systems where common-pool resource dynamics interact with
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8 household decisions.
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11 Within this framework, this study is designed to answer two research questions: (i) How do key
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13 economic parameters influence livelihood decisions of fishing households operating in an open-
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15 access common-pool resource (CPR) system? (ii) What policy measures could support both
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17 sustainable resource use and local well-being, given that these households depend on a fishery
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19 under an open access regime?
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23 To address these questions, we propose a Dynamic Fishing Household Production Model
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25 (DFHPM), following the household production model (HPM) tradition (Singh et al., 1986a).
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27 Household production models (HPMs) integrate production, consumption, and labor supply
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29 decisions within a recursive utility–profit maximization framework, recognizing that households
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31 both consume and produce goods, and that market and non-market goods are close substitutes. This
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33 perspective, which emphasizes the dual role of households in the economy, also draws attention to
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35 the frequent underestimation of their productive contributions in national accounts (Becker, 1990;
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37 Marszałek, 2025).
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43 Household production models are particularly well suited to SSF, as traditional models of fishing
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45 production typically assume profit-maximizing firms and abstract from self-consumption and the
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47 use of family labor. Likewise, standard consumption models do not account for how changes in
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49 prices or wages affect production and consumption decisions simultaneously.
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53 To our knowledge, no existing studies adapt the household production model framework to fishing
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55 households whose production depends on a common-pool resource under open-access regime. This
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57 constitutes a novel contribution of the paper.
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Our analysis yields several insights into how small-scale fishing households adjust their livelihood strategies when operating in a common-pool resource environment under an open-access regime. First, productive decisions, such as total harvest, fishing effort, and engagement in alternative activities, are highly elastic to changes in economic parameters, whereas consumption responses are generally less elastic. A key mechanism is the share of the catch allocated to self-consumption, that we will call γ , which acts as an adjustment margin that balances food security needs with income generation. Second, the model reveals that income, substitution, and profit effects interact in ways that generate non-trivial responses; for several parameters, the signs of the elasticities cannot be inferred theoretically and must instead be determined empirically. Third, the calibrated model indicates that improvements in external income opportunities—particularly higher wages outside the fishery—can simultaneously enhance household well-being and reduce pressure on fish stocks, in contrast to other policy levers, which tend to involve trade-offs between conservation and livelihoods. Finally, when accounting for stock dynamics, we find that declines in the resource can substantially weaken the effectiveness of policy interventions, suggesting that conservation measures must be paired with policies that protect household well-being. Together, these results provide a quantitative basis for designing socially inclusive and ecologically sustainable policy interventions for small-scale fishing communities.

This is a novel contribution to the discussion on how to reach sustainability of fish stocks, widely affected by overexploitation, while promoting socioeconomic development of fishing communities. Policies proposed either to improve fish stocks management or socioeconomic conditions of fishing communities separately might generate undesirable effects on conservation or communities' well-being and are less likely to be successful.

This analysis draws on rich socioeconomic information but lacks biological data on the fishery, a limitation common in many developing-country contexts where monitoring and research on

resource stocks are scarce. As a result, our simulations and conclusions focus on the socioeconomic dimensions of household behavior, while stock dynamics must be incorporated through informed assumptions rather than direct measurements.

This document is organized as follows. Following this introduction, we present a discussion on the use of a household production model within this framework. Section Three outlines our theoretical approach to the DFHPM, from which we derive supply and demand functions, along with their elasticities with respect to key parameters. Section Four describes the model's calibration and simulation, used to understand the responses of artisanal fishers in the context of a CPR under open-access. Section Five examines the effects of implementing different policies to manage fisheries or improve livelihood conditions, when dynamic changes in fish stock are taken into account. Section Six discusses the main findings, emphasizing the importance of policies that incorporate stock dynamics to enhance both community well-being and resource sustainability.

2. HOUSEHOLD PRODUCTION MODELS – HPM

2.1. Defining HPM

Rural households in semi-market economies produce goods and services that are partly for sale and partly for self-consumption. They consume market goods and inputs while allocating family labor for both subsistence production and market-oriented activities. One way to address the complex dynamics of rural households is by applying household production models (HPMs), which integrate production, consumption, and labor force supply decisions (Singh et al., 1986a). This perspective, which emphasizes the dual role of households in the economy, also highlights the frequent underestimation of their productive contributions in national accounts (Becker, 1990; Marszałek, 2025).

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4 Under a recursive structure, the HPM integrates utility and profit maximization (Singh et al.,
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6 1986a). HPMs are grounded in the idea that households are not only consumers of goods produced
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8 in the market economy, but also producers of goods for their own use. Within this framework,
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10 home-produced commodities and market goods are treated as close substitutes (Gronau, 1977;
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12 1980; Melmed-Sanjak & Santiago, 1996). Moreover, this class of models recognizes that goods
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14 produced and consumed within the households may carry values that differ from market prices—
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16 values shaped by cultural identity and social norms (Taylor & Adelman, 2003; Arslan & Taylor,
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18 2009; Arslan, 2011). In such settings, a portion of household production is intended not for market
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20 exchange but for self-consumption, including sharing with friends and neighbors. This distinction
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22 has proven useful in analyzing household demand—particularly for food—by differentiating
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24 between home-produced food and food consumed away from home (Hamermesh, 2007; Huffman,
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26 2010, 2011).

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33 HPMs examine the decision-making processes of rural households, whose behavior as producers
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35 influences their behavior as consumers and labor suppliers—and vice versa. These models help
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37 identify how government interventions (e.g., policies affecting input or output prices, decisions
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39 related to social investment projects, or conservation policies) shape household choices regarding
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41 how much to consume, how much to produce, how much labor to allocate to their main productive
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43 activity, and how much labor to supply to the market. These decisions, in turn, affect food security,
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45 income generation, and the diversification of productive activities within the household. A clear
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47 understanding of these dynamics contributes to more effective conservation planning (Shone &
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49 Caviglia-Harris, 2006; Guerrero, Albers & Langpap, 2025).

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55 HPMs differ from conventional consumer theory models. While the latter rely on substitution and
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57 income effects to predict household behavior in response to a change in the price of a good, HPMs
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59 also incorporate the profit effect (Singh et al., 1986b). As a result, outcomes in terms of
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4 consumption and labor supply in an HPM, may or may not align with those predicted by
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6 conventional models. For example, if the price of a good both produced and consumed by the
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8 household increases, a traditional consumer model would predict a decrease in the consumption
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10 due to the price rise. However, an HPM accounts for the profit effect, meaning that an increase in
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12 the price of the good also raises the household income and profits. This income gain can lead to
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14 higher demand for various goods, including those produced by the household itself. Thus, under
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16 an HPM, household demand for the self-produced good does not necessarily decline following a
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18 price increase. Instead, demand is "subject to two forces pushing in opposite directions" (Singh et
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20 al., 1986b). The net effect on consumption can therefore only be determined empirically, and on a
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22 case-by-case basis. With respect to production decisions, HPMs yield results that are consistent
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24 with those derived from conventional firm-theory models (Singh et al., 1986b).
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31 **2.2. Applying HPM TO FISHING HOUSEHOLDS**

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33 Artisanal fishers are, by definition, rural households embedded within a family farming or peasant
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35 economy (United Nations, 2018; Maldonado et al., 2007; Forero, 2003; Pérez & Pérez, 2002;
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37 Machado et al., 1993). Following Garay et al. (2009), these households are typically characterized
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39 by the following features: (i) land and capital are limiting factors—artisanal fishing households
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41 generally lack access to land, (ii) the family labor force is the primary factor of production; (iii) the
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43 output from extractive activities is used for both household self-consumption and market sale, (iv)
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45 households do not seek to maximize monetary profits but rather to secure food and income to meet
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47 basic needs and acquire other goods; (v) household income is low and derived from a range of
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49 productive activities; and (vi) households are integrated into the market through the sale of
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51 harvested products and labor, as well as through the purchase of inputs and other goods and
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53 services.
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Pascual-Fernández (1997) argues that fishers are more similar to hunters than to farmers, given the harvesting nature of the activity—where prey must be sought, located, and captured—and the absence of control over the resource or the natural systems in which the activity takes place (Smith, 1980; McCay & Ingold, 1988). Unlike farming, fishing relies on natural capital that is neither a private good nor, in most cases, collectively owned, but rather a common-pool resource (CPR), characterized by rivalry and non-excludability, in which allocation and enforcement of property rights are inherently complex (López-Martínez, Schriewer & Mesenguer-Sánchez, 2021).

Bearing this in mind, an adapted HPM could effectively explain the production and consumption decisions of fishing households. It can also be used to analyze how different policy interventions may influence those decisions and, ultimately, their well-being—particularly among fishing households with a profile of local wild harvesters living in remote, asset-poor, and less-developed villages, where dependence on fish resources is especially high (Wells et al., 2024; Guerrero, Albers & Langpap, 2025).

De la Montaña et al. (2015) follow the model proposed by Singh et al. (1986b) to examine how key economic parameters influence the hunting of wild fauna in indigenous communities of the Ecuadorian Amazon. In particular, this model develops an application of HPM for hunters, assuming that the resource—wildlife—is characterized by non-rivalry and non-excludability, that is a public good. De la Montaña et al. (2015) implicitly assume that harvest levels are sufficiently low relative to the rate of resource regeneration that neither the stock nor the profits of the hunting households are affected by the joint extraction of all hunters. Under this assumption, hunting can be treated as non-rival. Consequently, the model does not incorporate the impact of extraction on the resource stock.

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4 However, artisanal fisheries typically represent a classic case of a common-pool resource—
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6 characterized by rivalry—where harvest may exceed biological growth, leading to resource
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8 overexploitation under open-access conditions (Hardin, 1968; Gordon, 1954).
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11 Our fishing household production model builds on the frameworks developed by Singh et al.
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13 (1986b) and De la Montaña et al. (2015), but introduces a dynamic component in which household
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15 decisions depend on a fish stock that may vary exogenously due to aggregated fishing pressure or
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17 external factors such as climate change or water pollution. These stock variations influence both
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19 consumption and production decisions within the household.
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23 This dynamic model enables the analysis of how different policies may affect household well-being
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25 and exert pressure on fish stocks. It is particularly applicable to households engaged in the informal
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27 production of goods or services in the primary sector, which rely on natural resources that are
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29 difficult to exclude from use. It is also useful for analyzing policies that may influence both
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31 socioeconomic conditions and sustainability of the resource being used.
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34 35 36 37 **3. THE DYNAMIC FISHING HOUSEHOLD PRODUCTION MODEL – DFHPM**

38 39 40 **3.1. Definition of variables**

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42 In this model, the fishing household harvests a given species using a given fishing gear. The catch
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44 (Y) can either be consumed within the household or sold in the market. Let γ denote the proportion
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46 of the catch allocated to self-consumption; thus, γY represents household consumption, while
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48 $(1 - \gamma)Y$ corresponds to the quantity of fish sold. Although fishing is the household's primary
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50 productive activity, the household also engages in other activities to supplement its income. The
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52 household has a total amount of time available for productive activities $\bar{L}$, which can be allocated
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54 either to fishing L_y or to other income-generating activities L_{off} :
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$$\bar{L} = L_{off} + L_y \quad (1)$$

Household income is generated from three sources: the sale of fish, other productive activities, and government transfers. Income is therefore given by $P_y(1 - \gamma)Y + wL_{off} + T$, where w , P_y and T denote the wage rate earned in other income-generating activities, the price of fish, and the amount of transfers provided by the government, respectively. Household expenditures include the purchase of consumer goods, F , which may consist of other food items or provisions acquired in the market, at a price P_f . Additional costs arise from the fishing activity itself and depend directly on the quantity of fish extracted Y , and inversely on the stock of the resource in the sea, S , such that $C = C(Y, S)$. Therefore, total household expenses are: $P_f F + C(Y, S)$.

The fishing household is characterized by a utility function that depends on the self-consumption of fish (γY) and the consumption of other goods (F) as shown in Equation (2):

$$U = U(F, \gamma Y) \quad (2)$$

Household budget constraints require that total income equals total household expenditures; that is,

$$P_y(1 - \gamma)Y + wL_{off} + T = P_f F + C(Y, S) \quad (3)$$

In fisheries economics, it is typically assumed that the production function depends on two types of factors: those related to human activity—grouped under the term fishing effort, E —, and the natural factor, represented by the availability of the extracted resource or stock, S . Accordingly, the fishery production function can be expressed as $Y = Y(E, S)$.

Fishing effort (E) in turn, depends on the labor allocated to fishing, (L_y), and the capital invested, (K), such that $E = E(L_y, K)$. The production function can therefore be expressed as $Y = Y(L_y, K, S)$.

To capture the dynamic nature of the fish stock, we propose a two-period model [$t = 0, 1$], in which the stock evolves according to the natural ecological conditions, represented by the natural growth function $F(S_0)$, and the aggregated fishing activity, $\hat{Y} = \sum_i Y_i$:

$$S_1 - S_0 = F(S_0) - \hat{Y} \quad (4)$$

In this setting, the household maximizes the discounted sum of utility over the two periods, using a discount factor δ^t , subject to the discounted budget constraint, as well as the time endowment and production constraints:

$$\begin{aligned} \text{Max } \sum_{t=0}^1 \delta^t U_t(F_t, \gamma_t Y_t) &= \max u_0(F_0, \gamma_0 Y_0) + \delta u_1(F_1, \gamma_1 Y_1) \\ \text{Subject to } P_y(1 - \gamma_t)Y_t + wL_{offt} + T_t - P_f F_t - C(Y_t, S_t) &= 0 \\ Y_t &= Y(S_t, L_{yt}, K_t) \\ \bar{L} = L_{offt} + L_{yt} &\rightarrow L_{offt} = \bar{L} - L_{yt} \\ &\text{for } t = 0, 1 \end{aligned} \quad (5)$$

The term δ^t is defined as $\delta^t = \frac{1}{(1+r)^t}$, where r is the discount rate. Given this, the optimization requires:

$$\begin{aligned} \max_{F_i, Y_i, \gamma_i, L_{yi}} & u_0(F_0, \gamma_0 Y_0) + \delta u_1(F_1, \gamma_1 Y_1) \\ & + \lambda_0(T_0 + P_y(1 - \gamma_0)Y_0(\cdot) + w(\bar{L} - L_{y0}) - P_f F_0 - C(Y_0, S_0)) \\ & + \delta \lambda_1(T_1 + P_y(1 - \gamma_1)Y_1(\cdot) + w(\bar{L} - L_{y1}) - P_f F_1 - C(Y_1, S_1)) \end{aligned} \quad (6)$$

The level of harvest, the share of self-consumption, the consumption of other goods, and the allocation of time to fishing or other income-generating activities are endogenous variables that the household has to determine. In this model, prices are assumed to remain constant over time, and the household is not allowed to borrow or lend money between periods. As a result, decisions made

in period 0 are independent of those made in period 1. The only variable influencing intertemporal decisions is the change in the resource stock.

From the first-order conditions, two central equalities emerge. First, the marginal rate of substitution between market-purchased goods and self-consumed fish must equal the price ratio of these two goods, regardless of the period:

$$P_F/P_Y = MRS_{F_0, Y_0 Y_0} = MRS_{F_1, Y_1 Y_1} \quad (7)$$

Second, the value of the marginal productivity of labor must equal the external wage (w), regardless of the period:

$$\left(P_Y - \frac{\partial C}{\partial Y_0}\right) \frac{\partial Y_0}{\partial L_{Y_0}} = \left(P_Y - \frac{\partial C}{\partial Y_1}\right) \frac{\partial Y_1}{\partial L_{Y_1}} = w \quad (8)$$

To operationalize these expressions, specific functional forms are required for the utility function, the production function, and the cost function. For the utility function, we assume a Cobb-Douglas specification with a logarithmic transformation:

$$U = \ln(\hat{U}) = \alpha_f \ln F + \alpha_y \ln(\gamma Y) \quad (9)$$

Where parameters α_f and α_y represent the weight of the goods on household utility, and $\alpha_f + \alpha_y = 1$.

The production and cost functions must exhibit characteristics consistent with economic production theory. The Cobb-Douglas specification for the production function ensures factor substitutability and has demonstrated robust performance in simulation analyses (Yang, Chiang & Liu, 2022; Campbell, 1991):

$$Y = \tilde{\varphi} L_y^{\beta_1} K^{\beta_2} S \quad (10)$$

Where $\tilde{\varphi}$ and β_i represent productivity and the efficiency of fishing-associated factors, respectively. From the theoretical analysis, standard cost minimization using this type of functional form implies the following expression for the minimum cost function:

$$C(w, r, Y, S) = \left(\frac{Y}{\varphi S} \right)^{\frac{1}{\beta_1 + \beta_2}} \left(\frac{w^{\beta_1}}{r^{\beta_2}} \right)^{\frac{1}{\beta_1 + \beta_2}} \left[\left(\frac{\beta_1}{\beta_2} \right)^{\frac{\beta_2}{\beta_1 + \beta_2}} + \left(\frac{\beta_2}{\beta_1} \right)^{\frac{\beta_1}{\beta_1 + \beta_2}} \right] \quad (11)$$

$$= \left(\frac{Y}{\varphi S} \frac{w^{\beta_1}}{r^{\beta_2}} \right)^{\frac{1}{\beta_1 + \beta_2}} A$$

If the production function exhibits constant returns to scale in anthropic factors (L, K), the cost function takes the following form:

$$C(Y, S) = \frac{AY}{\tilde{\varphi} S} \frac{w^{\beta_1}}{r^{\beta_2}} = \frac{\theta Y}{S} \quad (12)$$

where θ reflects the intensity of cost function parameter ($\theta = \frac{A}{\tilde{\varphi}} \frac{w^{\beta_1}}{r^{\beta_2}}$) while θ/S reflects the marginal and the average cost of the catch.

Data from the survey demonstrated that credit markets are highly restricted in the context of our analysis and that capital (represented by vessels, engines, and gear) behaves more as a fixed factor. Therefore, we assume that capital is fixed in this model, and thus, the production function can be written as a function of labor allocated to fishing, the resource stock, and technological parameters:

$$Y = \tilde{\varphi} \bar{K}^{1-\beta} L_y^\beta S = \varphi L_y^\beta S \quad (13)$$

Notice that $\tilde{\varphi} \bar{K}^{1-\beta} = \varphi$. Using this assumption about the utility and production functional forms, we are able to estimate the household's decision functions related to consumption and production.

3.2. Household Supply and Demand Functions

The solution to the model yields the functions of labor supply for fishing (L_y), labor supply for non-fishing productive activities (L_{off}), fish catch (Y), proportion of fish for household

consumption (γ), quantity of fish consumed within the household (γY), and demand for goods purchased on the market (F), for each period. As a result of temporal independence, these functions share the same functional form across periods, but their value depends on the contemporaneous fish stock. Estimated functions are presented in Table 1. As expected, production variables (fish supply and labor supply for fishing and non-fishing activities) behave as conventional models would predict. For instance, the total fish catch—Equation (16)—will depend directly on the resource stock (S), the productivity of the fishing-associated factors (φ), the efficiency of the fishing labor force (β), and the net profit per unit of fish ($SP_y - \theta$), and inversely on the wage received from other productive activities (w).

Consumption variables (self-consumed fish and goods purchased on the market) exhibit the particular characteristic structure of an HPM. The quantity of fish caught that the household allocates for self-consumption (γY)—Equation (18)—depends directly on the weight of fish consumption for household utility (α_y), on the time available for work ($\bar{L}$) and the transfers (T), on the productivity of fishing φ , and on the marginal net benefit of fishing ($SP_y - \theta$). The signs of the effects of the fish price and wages, however, are not theoretically unambiguous. This ambiguity arises because fish price plays a dual role. On one hand, it has a negative effect by reducing demand for the good; on the other, it has a positive effect through the profit channel, as higher prices increase income from fish sales, thereby expanding the household's capacity for consumption.

A similar mechanism operates for wages. On the one hand, the income from alternative activities increases along with wages, leading to greater consumption of goods, including fish. On the other hand, a higher wage reduces the incentive to fish, thereby decreasing the quantity of fish caught and available for consumption.

Table 1. Summary of the optimal supply and demand functions resulting from the DFHPM. As the functions exhibit the same functional form in both periods, we omit the sub index t .

Demand/ Supply	Mathematical expression	Eq.
L_y = Fishing labor supply	$L_y = \left(\frac{\beta \varphi S}{w} \left(P_y - \frac{\theta}{s} \right) \right)^{\frac{1}{1-\beta}}$	(14)
L_{off} = Non-fishing labor supply	$L_{off} = \bar{L} - \left(\frac{\beta \varphi S}{w} \left(P_y - \frac{\theta}{s} \right) \right)^{\frac{1}{1-\beta}}$	(15)
Y = Fishcatch	$Y = \left(\varphi S \left(\frac{\beta}{w} \left(P_y - \frac{\theta}{s} \right) \right)^{\beta} \right)^{1/1-\beta}$	(16)
γ = Proportion of fish for household consumption	$\gamma = \frac{\alpha_y}{P_y} \left[\bar{L} \left(\frac{w}{\varphi S \left(\beta \left(P_y - \frac{\theta}{s} \right) \right)^{\beta}} \right)^{1/1-\beta} + \left(P_y - \frac{\theta}{s} \right) (1 - \beta) \right]$	(17)
γY = Demand for fish for household consumption	$\gamma Y = \frac{\alpha_y}{P_y} \left[w \bar{L} + (1 - \beta) \left(\left(\frac{\beta}{w} \right)^{\beta} \varphi S \left(P_y - \frac{\theta}{s} \right) \right)^{1/1-\beta} \right]$	(18)
F = Demand for other consumption goods	$F = \frac{\alpha_f}{P_f} \left[w \bar{L} + (1 - \beta) \left(\left(\frac{\beta}{w} \right)^{\beta} S \varphi \left(P_y - \frac{\theta}{s} \right) \right)^{1/1-\beta} \right]$	(19)

Assumptions: $(SP_y - \theta) \geq 0$; $0 < \beta < 1$.

Note that resource availability plays a vital role in household well-being, as it enhances the household's ability to consume both fish and other goods.

3.3. Comparative statics: Elasticities

Using the functions estimated by the model (L_y , L_{off} , F , Y , γ , γY), we calculated the elasticities of these endogenous variables in response to changes in several key economic parameters: prices (P_f , P_y , w), technical and cost parameters of the fishing activity (φ , θ), direct transfers (T), the

households' time availability for productive activities ($\bar{L}$), and the availability of the extracted resource (S). These are presented in SM1. In some cases, the sign of elasticities is unequivocal; in others, it is ambiguous, and the final value depends on the context. As discussed above, some parameters influence variables in opposite directions, so the final sign will depend on the relative strength of each. Similarly, the magnitude of the elasticities is an empirical matter, determined based on household survey data. Table 2 summarizes the signs of the theoretical elasticities derived based on the DFHPM. Among the various parameters, it must be noted that the available resource stock plays a key role in household decision-making. Higher fish stocks lead to a greater time allocation to fishing (and reduced time spent on other activities), larger harvests, and higher levels of fish self-consumption. The additional income generated from larger fish harvests also enables increased consumption of other goods.

Table 2. Signs of derived theoretical elasticities in response to changes in key parameters

Variable	P_Y	P_F	w	θ	ϕ	T	$\bar{L}$	S
L_y	+	0	-	-	+	0	0	+
L_{off}	-	0	+	+	-	?	+	-
Y	+	0	-	-	+	0	0	+
γ	?	0	+	?	-	+	+	?
γY	?	0	?	-	+	+	+	+
F	+	-	+	-	+	+	+	+

A key distinction of the DFHPM relative to classical consumer-theory-based models is the profit effect, which arises when an increase in the price of a good produced by the household directly raises its total income. For small-scale fisheries, higher fish prices both increase household income

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4 and influence household decisions in two important ways: (i) consumption of self-produced fish
5 increases, and (ii) labor allocation adjusts as households devote more time to fishing—substituting
6 labor away from other activities.
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11 The variable γ , representing the share of the catch allocated to self-consumption, is central in
12 mediating these effects, linking production and consumption within the household and determining
13 the direction and magnitude of the profit effect. Its inclusion enables the model to capture
14 behavioral responses that would not be predicted by traditional consumer-only models, providing
15 a more nuanced understanding of household decision-making in small-scale fisheries.
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23 However, variable γ , which represents the proportion of fish allocated to self-consumption, has a
24 theoretically indeterminate sign. This reflects the inherent trade-off in the household production
25 model: a larger harvest leads to increased fish consumption, but it also provides more opportunities
26 to generate income through market sales. As a result, the final value is context-specific and
27 empirically defined.
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35 Empirically, with respect to the allocation between self-consumption and sales, our data show that,
36 on average, households catch 56.3 kg of fish per week, of which 86% is sold and 13% is reserved
37 for household consumption. Interestingly, although total catch and the portion sold fluctuate
38 throughout the 12-month data collection period, the amount reserved for self-consumption remains
39 almost constant at around 7 kg per week. Considering the average household size of 3.9 members,
40 this implies that each household member consumes approximately 0.25 kg of fish per day from the
41 portion allocated to self-consumption, more than 90 kg per year per capita.
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52 This stable consumption highlights the role of self-produced fish as a buffer against market and
53 ecological variability, providing food security, which demonstrates the relevance of using the
54 DFHPM framework. While the profit effect may be small on average, it is particularly relevant in
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households with higher reliance on self-consumption, demonstrating the non-standard cases where the model captures dynamics that classical consumer theory would miss.

4. MODEL SIMULATION

4.1. Overview of the Barú Artisanal Fishery

The Barú artisanal fishery is a small-scale, coastal, multispecies fishery that targets reef-, mangrove-, and nearshore-associated species. Local monitoring documents high species diversity (≈ 80 species) with snappers (*Lutjanus* spp.), octopus, barracuda, spiny lobster, and jacks among the main components of landings (Viloria et al. 2015). Fishing effort is concentrated in two landing sites (*Pesbarú* and *Tuntuneco*). The dominant gears—handlines, breath-hold diving, and *nasas*—are relatively selective, and local norms discourage the use of ecologically harmful gears (Villamil Medrano et al. 2020).

The fishery primarily exploits resident coastal stocks with limited migratory behavior and operates within an open access context, partly influenced by the zoning of the Corales del Rosario and San Bernardo National Natural Park. Seasonal variability in catch and effort, heterogeneous CPUE across gears and sites, and signals of moderate-to-high overexploitation risk have been documented (Viloria et al. 2015; Maldonado et al. 2022). This brief overview provides the ecological and institutional context for interpreting the model simulations.

4.2. Data

The theoretical model is empirically validated by quantifying its key parameters using data collected from fishing households in the village of Barú. We classify a household as a *fishing household* when the head of household's primary economic activity is or has historically been fishing, and when the household is also recognized by the community as belonging to the fishing sector. This category might include households that have traditionally depended on fishing but may

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4 currently be transitioning into other activities or those that may have stopped fishing due to the
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6 advanced age of the household head. Conversely, a *non-fishing household* is defined as one in
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8 which the head of household's main economic activity is not related to fishing.
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11 According to a household census conducted between June and July of 2018, the village comprised
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13 801 households, of which 158 were identified as fishing households. From this population, we
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15 randomly selected a stratified sample of 97 fishing households—sufficient to achieve a confidence
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17 level above 95% with a 5% error margin. A set of monthly surveys was designed and administered
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19 between July 2018 and October 2019.
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23 Fishing households in Barú typically consist of four members, with an average age of 26 years.
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25 These households have an illiteracy rate of 14%, which rises to 22% when considering only heads
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27 of household. On average, household heads have 4.7 years of formal education. In terms of labor
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29 diversification, fishing households in Barú report an average of 1.71 income-generating activities
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31 per household. The most common primary occupations among household heads include fishing,
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33 maritime transport, fish storage, fish trading, and rental of fishing equipment. Regarding fishing
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35 specifically, 100% of surveyed fishers were men, with an average age of 46 years, and an average
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37 of 4.3 years of education.
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43 The most commonly used fishing methods are handlining and diving. When fishing is the primary
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45 activity, 47% of fishing households prefer handlining, while 40% opt for free diving. A key
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47 characteristic of fishing is its flexibility—it can serve as either a primary or secondary activity and
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49 may be practiced on a full or part-time basis. Fishing plays a vital role in generating income, not
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51 only for households that rely on it as their main source of livelihood, but also as a coping strategy
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53 for those who engage in it as a secondary activity during times of hardship. On average, 85% of
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55 the catch is sold, 13% is consumed by the household, and 2% is gifted to other households.
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In Barú, the average monthly monetary expenditure of fishing households is approximately USD 836 (PPP¹), while the average per capita household expenditure is USD 229 (PPP). The total average monthly income of these households is around USD 1,233 (PPP). Fishing contributes directly to 37% of household income and indirectly to an additional 5.8%. According to the headcount poverty index, 27.1% of fishing households in Barú fall below the national poverty line, and 4.5% are below the extreme-poverty line.

4.3. Simulation strategy

The simulation procedure follows three steps. First, we use the household survey to obtain empirical approximations of the model's structural parameters, relying on different calibration and estimation approaches—depending on data availability—as described in Section 4.4. Second, we input these calibrated parameters into the theoretical household production model to compute the predicted values of the endogenous variables, such as fishing effort, total harvest, and the share of the catch allocated to self-consumption. Third, we compare these model-implied outcomes with their empirical counterparts in the survey data, looking to find the closest correspondence between predicted computed values and observed variables. This exercise allows us to assess the model's internal validity and to evaluate how well its predicted behavioral responses align with observed household decisions.

4.4. Model Parameters

The FHPM includes a set of parameters that require calibration so that when incorporated into the model's functions, the resulting predicted values closely match the observed data. These parameters can be grouped into the following categories:

¹ Purchasing power parity of USD to COP for 2019: 1,430 COP / 1 USD.

- **Market prices.** In this model, prices include fish price, P_y , wage rate, w , and price of other goods, P_F . Fish price was estimated based on survey data at USD 11.73 (± 5.57) PPP per kilogram, and wage for other activities at USD 2.52 (± 3.94) PPP per hour. The value of P_F was set at 1, assuming it is a numeraire good.
- **Household characteristics.** Two parameters are needed to calibrate the model. The first is the total amount of time available to the household for work. Based on the survey data, we estimated this value $\bar{L}$, at 70 hours per week (69.98 ± 37.64). The second parameter is government transfers, also obtained from the survey, which were estimated at USD 18.44 (PPP).
- **Production and cost function parameters.** As mentioned earlier, survey data showed that credit markets are highly restricted in this setting and that physical capital (represented by vessels, engines, and gear) is not a variable factor. Therefore, we assume that capital is fixed in this model, and thus, the production function can be written as:

$$Y = \tilde{\varphi} L_y^\beta \bar{K}^{1-\beta} S = \varphi L_y^\beta S \quad (20)$$

We define $\tilde{\varphi} \bar{K}^{1-\beta} = \varphi$. Thus, its estimation requires the calculation of parameters φ and β . Cost function, in its reduced form can be written as $C = \Theta Y/S$, meaning that we need an estimate of parameter θ . To approach the parameters associated with the production and cost functions, the full survey dataset can be used to econometrically estimate the parameters corresponding to the adopted functional forms. For the production function, the estimated model is:

$$\ln(Y_{it} + 1) = \varphi + \beta \ln(L_y) + u_{it} \quad (21)$$

As for the cost function, a linear functional form is estimated:

$$C_{it} = \alpha_0 + \alpha_1 Y_{it} + u_{it} \quad (22)$$

Where α_0 is a constant that captures the time-invariant factors affecting the household's cost of fishing, which presumably correspond to fixed costs. This term does not represent a structural fixed cost; rather, it is the empirical counterpart of the theoretical cost parameter under a common-stock assumption. No fixed costs are being introduced relative to the model. Parameter α_1 reflects the term θ/S . Note that the resource stock is an unobservable variable in the model, and, as such, it cannot be separately identified in the econometric estimation. u_{it} is the error term.

In the theoretical model, the resource stock S directly increases the marginal productivity of fishing effort and reduces marginal costs. However, no spatially disaggregated or time-varying measures of stock abundance are available for our study area. As a result, equations (21) and (22) are estimated in reduced form, where the influence of S enters the composite error term u_{it} . Specifically, unobserved variation in stock abundance affects the level of output (Equation 21) and costs (Equation 22), but cannot be controlled for explicitly.

Thus, the estimated parameters β and α_1 should not be interpreted as structural or causal elasticities. Their role is instead to provide empirical magnitudes that are consistent with, and used to calibrate, the dynamic household model developed in Section 2. In principle, stock dynamics could be incorporated empirically through habitat indicators, lagged CPUE, or effort-based proxies, but such data are not available in our setting. We therefore make it explicit that this empirical component is a reduced-form calibration compatible with the theoretical structure.

Additionally, because the stock S is unobserved and common to all households operating in the same fishery, it cannot be included directly in the estimation. In practice, S is absorbed into

the constant term of the log-linearized specification, and any household-specific deviations from the common stock level enter the composite error term.

- **Unobservable parameters.** Some parameters are not directly observable from the data. One such set relates to the utility function parameters. In our model, we assume that $\alpha_y + \alpha_f = 1$. For a Cobb-Douglas utility function, the resulting Marshallian demand functions take a standard form, generally expressed as $X = \alpha_x M / P_X$, where M denotes available income, and P_X the price of the good. From this relationship, we can derive that $\alpha_x = P_X X / M$. Applying this approach to our utility function and goods, we can assume that:

$$\alpha_{YY} = \frac{P_Y \gamma Y}{M}; \quad \alpha_F = \frac{P_F F}{M} \quad (23)$$

These parameters represent the proportion of household income allocated to the expenditure on each good. Their values can be inferred from the survey data.

The resource stock, S , is not directly observable from the survey or any other available data sources, which leads us to assume a normalized value of 1, for calibration purposes. This normalization does not imply a constant stock over time. The model can be equivalently expressed in terms of a normalized stock $\tilde{S}_t = S_t / S_0$, where only changes in $\tilde{S}_t$ matter for household decisions. In the dynamic exercise, we start from a value for the stock, $\tilde{S}_0 = 1$.

Assuming that aggregate extraction exceeds stock growth, we define $\tilde{S}_1 < 1$ as a reduction in the stock between the two periods. Furthermore, because we lack aggregate biological information on the initial stock level, total fishing effort, and the stock growth function, the dynamic exercise cannot quantify the actual change in stock resulting from a given policy. Instead, we simulate a small (1%) reduction in the stock to illustrate the model's behavioral implications under policy-induced changes in resource scarcity.

4.5. Model Calibration

To identify the parameter values that best approximate the observed data, we apply a numerical calibration method. Table 3 reports the estimated values used for calibrating the supply and demand system, as well as the resulting calibrated parameters. Given the uncertainty surrounding some parameter values, we present intervals that we define as feasible for the analysis. Then, we report 90% confidence intervals for the estimated values.

4.6. Household Supply, Demand, and Utility Functions

Table 4 presents the results for supply, demand, and utility functions in the FHPM, based on the calibrated parameters. The estimated values closely match the observed values, consistently falling within 0.2 standard deviations. This level of accuracy provides confidence that the parameters and computed endogenous variables reliably reflect the behavior of fishing households in Barú. We also present 90% confidence intervals for the estimated values.

While we also compare the calibrated values and the resulting model-generated outcomes with the observed means and medians, this exercise should be understood as a basic internal consistency check rather than a formal validation procedure. In other words, the “validation” conducted here is limited to assessing the empirical plausibility of the calibrated parameters and ensuring that the model reproduces realistic empirical magnitudes. This clarification helps align the terminology with the actual scope of the empirical analysis.

Table 3 Estimated parameters for the FHPM and final calibrated value

Variable/ parameter	Symbol	Estimated interval (Lower and upper bounds)	Approach	Value used in calibration (C.I.)
Weight of consumption of other goods for household utility	α_f	0.75-0.95	Marshallian demand function	0.86 (0.82 0.90)
Weight of fish self-consumption for household utility	α_y	0.05- 0.25		0.14 (0.13 0.15)
Total fishing-factor productivity	φ	2.358 – 3.628	Production function	3.44 (3.27 3.61)
Labor efficiency in fishing	β	0.469 – 0.595	regression	0.5 (0.48 0.53)
Cost parameter	θ	2.763 – 3.470	Cost function regression	3.14 (2.98 3,29)
Fish market price	P_y	11.14-12.32	Average of observed values	11.73 (11.14 12.32)
Hours allocated to work by the household	$\bar{L}$	66.50-73.50	Average of observed values	70 (66.5 73.5)

Table 4 Simulation of results for supply, demand, and utility functions

	Symbol	Observed data		FHPM
		Mean (s.d.)	Median	Calibrated parameters (90% C.I.)
Labor supply for fishing (hours per week)	L_y	35.43 (18.54)	35	34.38 (26.2 44.7)
Labor supply for other productive activities (hours per week)	L_{off}	33.10 (34.52)	32	35.62 (15.8 50.4)
Fish catch (kilograms per week)	Y	20.80 (21.42)	15	20.16 (10.1 – 31.5)
Proportion of fish caught for self-consumption (percentage)	γ	0.18 (0.17)	0.14	0.17 (0.09 0.26)
Quantity of self-consumed fish (kilograms per week)	γY	3.72 (2.29)	2.50	3.36 (2.58 4.87)
Market demand for food (USD PPP per week)	F	205.87 (175.74)	176	242.06 (99.4 312.3)

4.7. Estimated Elasticities

Table 5 reports the supply and demand elasticities in the FHPM with respect to the estimated parameters. Yellow cells indicate variables that exhibit an elastic relationship with the parameters, purple cells indicate inelastic relationships, and green cells denote unitary elasticity. The theoretically expected sign is reported in parentheses.

Table 5 Value of estimated elasticities. The theoretically expected sign in parentheses

Variable	P_Y	P_f	w	θ	φ	T	$\bar{L}$	S
L_Y	+2.73 (+)	0 (0)	-2.00 (-)	-0.73 (-)	+2.00 (+)	0 (0)	0 (0)	+2.73 (+)
L_{off}	-2.63 (-)	0 (0)	+1.93 (+)	+0.70 (+)	-1.93 (-)	0 (0)	+1.96 (+)	-2.63 (-)
Y	+1.37 (+)	0 (0)	-1.000 (-)	-0.365 (-)	+2.00 (+)	0 (0)	0 (0)	+2.37 (+)
y	-1.47 (?)	0 (0)	+1.34 (+)	+0.125 (?)	-1.34 (-)	+0.07 (+)	+0.67 (+)	-1.47 (?)
γY	-0.10 (?)	0 (0)	+0.34 (?)	-0.24 (-)	+0.66 (+)	+0.07 (+)	+0.67 (+)	+0.90 (+)
F	+0.90 (+)	-1.000 (-)	+0.34 (+)	-0.24 (-)	+0.66 (+)	+0.07 (+)	+0.67 (+)	+0.90 (+)

The estimated elasticities are consistent with theoretical predictions in terms of their signs. In cases where the sign of elasticities is theoretically ambiguous, the following results are obtained. Specifically, a negative sign is found for the elasticity of fish consumption within the household (and the proportion of fish allocated to self-consumption) in relation to fish price.

Recall that the FHPM incorporates two opposing forces. On the one hand, as a consumer, the household reduces consumption in response to price increases. On the other hand, as a producer, an increase in price raises household income, which in turn allows for greater consumption of goods—implying a positive effect. In this case, the estimated elasticity is negative, indicating that

the standard consumer effect dominates the profit effect, despite an increase in the quantity of fish harvested. As shown in the table, the parameter γ is elastic and negatively related to the price of fish, outweighing the effect on the harvesting.

An increase in the wage rate for non-fishing activities has two opposite effects. On the one hand, household income rises as a result of greater labor allocation to these alternative activities. On the other, a reduction in time allocated to fishing reduces the catch, which in turn reduces income from fish sales. Consumption is expected to increase in the former case and to decrease in the latter. The results indicate that the income gained from alternative work more than offsets the income lost due to lower fish sales. This net effect leads to an increase in the demand for both self-consumed fish and other goods. Faced with a reduced catch resulting from higher wages, fishers choose to allocate a larger share of their catch to self-consumption. This result is crucial for policy analysis.

As expected, higher fishing costs reduce the effort devoted to fishing and, consequently, the catch. This, in turn, lowers household income and leads to a reduction in the consumption of goods. Although the proportion of fish allocated to self-consumption increases, total household fish consumption still declines due to reduced fishing activity.

Changes in lump-sum transfers do not affect fishing decisions. However, they do affect consumption decisions, by increasing them, albeit with very low elasticity—likely reflecting the relatively small size of transfers received by the population in Barú.

An increase in the household's available labor force (e.g., through the arrival of new members capable of working) tends to be allocated to alternative income-generating activities, thereby increasing household income. This additional income leads to higher consumption of all goods, including fish for self-consumption, implying that a larger proportion of the catch is retained for the household.

Finally, an increase in the resource stock induces fishers to raise their fishing effort, resulting in higher catches and enabling greater consumption of both fish and other goods within the household, even if the proportion of the catch allocated to self-consumption decreases. This is consistent with the analysis presented in SM2, which shows that a 1% reduction in fish stocks in South America has led to a 2.5% decrease in catches.

4.8. Intertemporal effects

Households make decisions individually based on their private information and cannot observe the resource stock or its changes. This represents the typical externality associated with the exploitation of a common-pool resource. Although the stock is not directly observable at the individual level, it is a common-pool resource and individual decisions aggregate to affect overall resource availability. Under open access, if fishing households perceive positive benefits from fishing, effort will increase in order to capture these rents. Household profits from fishing are defined by the term $(P_y - \theta/S)$, which denotes the marginal benefits. As long as this term remains positive, fishing effort will continue to increase.

The aggregate behavior of the stock will respond to the evolution equation:

$$\begin{aligned} S_1 &= S_0 + F(S_0) - \hat{Y} \\ S_1 - S_0 &= \frac{\Delta S}{\Delta t} = F(S_0) - \hat{Y} \end{aligned} \tag{24}$$

Where $\hat{Y}$ refers to the aggregated harvest of all participating fishing households and $F(S)$ denotes the resource growth function. If $\frac{\Delta S}{\Delta t} < 0$ the stock will decrease in period 1, and therefore, households will adjust their decisions accordingly.

These dynamics also help interpret how the system adjusts over time when economic conditions change. If fish prices rise, the marginal and average benefits of fishing $(P_y - \frac{\theta}{S})$ increase, leading

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4 to a persistent expansion of effort. Under open-access conditions, these higher returns would also
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6 attract additional users into the fishery, accelerating resource depletion. As long as profits remain
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8 positive, additional effort will be applied to the fishery, which in turn reduces the fish stock. A
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10 lower stock will reduce profits. In our model, the long-run open-access equilibrium stock
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12 corresponds to $S^{OA} = 0.2674$, which is substantially below the reference stock $S_0 = 1$.
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15 This behavioral response is consistent with the literature showing that common-pool resources
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17 often act as informal insurance mechanisms for low-income households, who adjust effort and
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19 extraction in response to negative shocks (Baland & Francois, 2005). Noack et al. (2018) also show
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21 that households adjust internal allocation rules when facing resource scarcity, emphasizing how
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23 preferences and constraints jointly determine consumption–production trade-offs. This aligns with
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25 our finding that a reduction in the stock induces changes in γ and other household decisions as they
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27 reoptimize under tighter constraints. Table 5 illustrates that a reduction in S , will lead to a decrease
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29 in harvest, time devoted to fishing, self-consumption, and consumption of other goods, while time
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31 allocated to other activities, and share of self-consumption increase.
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35 According to FAO (2024), sustainable stocks have diminished from 90% in 1974 to 62.3% in 2021,
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37 implying an annual reduction rate of 0.59%. We performed a RAM Legacy Stock Assessment
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39 Database (see SM2) analysis and found that, in South America, fish stocks have reduced by 52.6%
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41 from 1990 to 2015, and 36.86% in the 2000-2016 period, implying annual reduction rates greater
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43 than 2%. Global data reveal similar trends. Using a conservative estimate of a 0.6% annual decline
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45 we simulate changes in fish stock for 10, 15, and 20 years. The results in Table 6 are based on the
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47 outcomes from the last column of Table 5. As fish stocks decline, the labor dedicated to fishing
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49 drops significantly and is reallocated to alternative activities. Although labor markets are imperfect,
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51 human capital is limited, and other constraints are present, it is assumed that fishers leaving the
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53 sector are able to find employment in other activities. Fish catches also decrease at a more than
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proportional rate. Nevertheless, fishers choose to increase the share of the catch allocated to self-consumption. Despite this adjustment, overall consumption of both fish and of other goods declines. This outcome reflects a reduction in income from fish sales, which forces fishers to cut back on total consumption.

Table 6 Effect of changes in fish stock on model variables

Variable	Estimated	Stock	Stock	Stock
	elasticities	change	change	change
		-11.88%	-8.84%	-5.89%
		20 years	15 years	10 years
Fishing labor (Ly)	+2.7	-32.2%	-24.1	-16.1%
Alternative labor (Loff)	-2.6	+31.0%	+23.3	15.5%
Harvest (Y)	+2.4	-27.9%	-20.9%	-13.9%
Share of harvest	-1.5	17.3%	13,0%	8.6%
consumption (γ)				
Self-consumption (γY)	+0.9	-10.6%	-7.9%	-5.3%
Other consumption (F)	+0.9	-10.6%	-7.9%	-5.3%
Well-being (utility)	+0.9	-10.6%	-7.9%	-5.3%

5. POLICY INTERVENTIONS

FAO declared 2022 as the Year of Artisanal Fisheries and Aquaculture and proposed a Global Action Plan to empower small-scale artisanal fisheries and secure a sustainable future for the sector (FAO, 2021). As fish stocks are affected by overexploitation, the well-being of fishing communities declines as catches fall, reducing both self-consumption and sales and, in turn, limiting the consumption of other goods. In line with FAO's Global Action Plan, Vargas-Morales

(2023) compiled a set of policies designed to support fishing communities, which can be simulated using our model. These policies include increases in fish prices, improvements in fishing technology, subsidies for fishing, lump-sum transfers to households, and higher remuneration for alternative activities. The theoretical model allows us to simulate the effect of these policies. In doing so, we can examine the relationship between impacts on living standards and the pressure exerted on resources under different policy options. This approach is novel in that it enables policymakers to observe the joint effect of policies on both conservation and household well-being. The first policy we analyze is an increase in **fish prices** (P_y), which may occur as a result of various exogenous changes. For example, increased demand for fish driven by policies that promote beach tourism could raise fish prices. Note that a price increase simulated here is not necessarily associated with any management strategy aimed at promoting sustainable fishery use.

Given the model assumption of a single representative species with a single price, the possibility that consumers might shift toward consuming more sustainable, cheaper, or smaller fish is not considered. The simulation results (Table 7) show that—in gross terms—this policy would increase the catch (see column Direct effect), placing additional pressure on the resource stock. Although higher prices would also increase fishing households' well-being in terms of income and utility, the price increase would reduce fish self-consumption. Alderman (1986) argues that higher income and less self-consumption may lead to increased consumption of staple food such as grains, which contain lower-quality protein than fish. Therefore, larger catches and income do not necessarily translate into improved food security. Kawarazuka and Béné (2010) highlight the risk that food-insecure households may opt to sell a larger proportion of their fish catch in order to secure minimum quantities of basic goods, often perceived as a priority. In such cases, food consumption may come at the expense of nutritional security.

Table 7 Effect of selected policies on key model variables. “Direct effect” refers to outcomes without accounting for stock changes, while “after stock reduction” reflects the effects following a 1% decline in stock levels. Values are expressed as percentages.

Variable	Increase 1% in fish prices		Improved 1% fishing technology		1% fishing costs subsidy		1% increase in transfers		Alternative income, 1% increase wage	
	Direct	After	Direct	After	Direct	After	Direct	After	Direct	After
	effect	Stock reduction	effect	Stock reduction	effect	Stock reduction	effect	Stock reduction	effect	Stock reduction
L_Y	2.73	-0.07	2.00	-0.78	0.73	-2.02	0	-2.73	-2.00	-4.68
L_{off}	-2.63	-0.07	-1.93	0.65	-0.70	1.91	0	2.63	1.93	4.61
Y	1.36	-1.03	2.00	-0.41	0.36	-2.01	0	-2.36	-1.00	-3.34
y	-1.47	-0.02	-1.34	0.10	-0.12	1.34	0.07	1.53	1.34	2.83
yY	-0.10	-1.00	0.66	-0.25	0.24	-0.66	0.07	-0.83	0.34	-0.56
F	0.90	-0.01	0.66	-0.25	0.24	-0.66	0.07	-0.83	0.34	-0.56
U	0.76	-0.15%	0.66	-0.25	0.24	-0.66	0.07	-0.83	0.34	-0.56
Change needed to overcome 1% stock reduction										
	1.20%		1.38%		3.77%		13.85%		2.66%	

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4 However, if fish stocks are affected by overexploitation, the effectiveness of this policy may
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6 quickly diminish. Using as an example, a reduction of 1% in fish stock, as shown in Table 7
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8 (column *After stock reduction*), a 1% increase in fish prices is not sufficient to restore catch levels
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10 to those observed prior to the stock decline.
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14 We estimate that a 1.20% increase in fish prices is required to restore household well-being to the
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16 level observed prior to a 1% reduction in stock. That is, the compensating variation associated with
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18 each 1% decline in stock corresponds to a 1.20% increase in fish prices.
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21 A second policy involves improvements in *fishing technology* (φ), which increase catch
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23 productivity and, in turn, raise fishing effort. In the model, this parameter also captures capital
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25 assets such as vessels, gear, and fishing equipment. The simulation shows that the catch of fishers
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27 who adopt improved technology increases more than proportionally, thereby exerting additional
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29 pressure on the resource stock. In this context, we do not consider technologies designed to promote
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31 sustainability but rather those that enhance overall fishing productivity. Examples of such policies
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33 include the provision of larger vessels or more effective fishing gear. The resulting increase in
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35 catch translates into higher income. Although the proportion of the catch allocated to self-
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37 consumption declines, its absolute quantity rises, thereby improving households' intake of
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39 nutritional food.
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45 Once the stock is affected, the impact of this policy may also be undermined. A 1% decrease in
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47 stock reduces the policy's effect, resulting in a 0.41% decline in fishing activity. The overall impact
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49 on household well-being is similarly affected: what was initially a 0.66% increase becomes a
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51 negative outcome of -0.25% following a 1% reduction in stock.
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54 The compensating variation required to offset the stock decline and its effects on well-being would
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56 therefore involve a 1.38% increase in the technology parameter to compensate for a 1% reduction
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58 in the stock.
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4 The third intervention involves reducing **fishing costs** (θ) through subsidies or lower prices for
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6 variable inputs, such as oil. In the simulation, this policy generates effects similar to those of
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8 technological improvements, although the magnitude of its impact across all analyzed variables is
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10 notably smaller. Once the stock effect is taken into account, fishing activity declines by more than
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12 2% for each 1% reduction in stock, while consumption of fish and other goods falls by 0.66%,
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14 along with overall well-being. To offset the impact of a 1% decrease in stock on well-being, fishing
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16 costs would need to be reduced by 3.77%.
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21 A fourth possible policy is to increase **direct transfers** (T), whether conditional or unconditional
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23 lump-sum transfers to households. These transfers do not affect catch levels or the allocation of
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25 time between fishing and other activities; they influence only consumption decisions. Given the
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27 relatively low value of current transfers, however, a 1% increase has only a modest impact.
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29 According to the model, compensating for a reduction in stock would require an increase in direct
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31 transfers of approximately 14%. At the same time, this magnitude serves as a proxy for the
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33 importance of fish stocks in determining household well-being.
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38 The fifth policy option involves strengthening **alternative income-generating activities**—such as
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40 independent productive projects or employment outside fishing—that provide income for the
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42 household. In the model, this policy is simulated as an increase in the wage rate, w . This
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44 intervention has a positive effect on household well-being by increasing income and consumption,
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46 reducing harvest levels, and therefore alleviating the pressure on the fishing resource. It also leads
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48 to increased household self-consumption of fish. Although income from fishing declines
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50 substantially, income from alternative activities increases more than proportionally, ensuring that
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52 total household income does not fall and that overall well-being improves.
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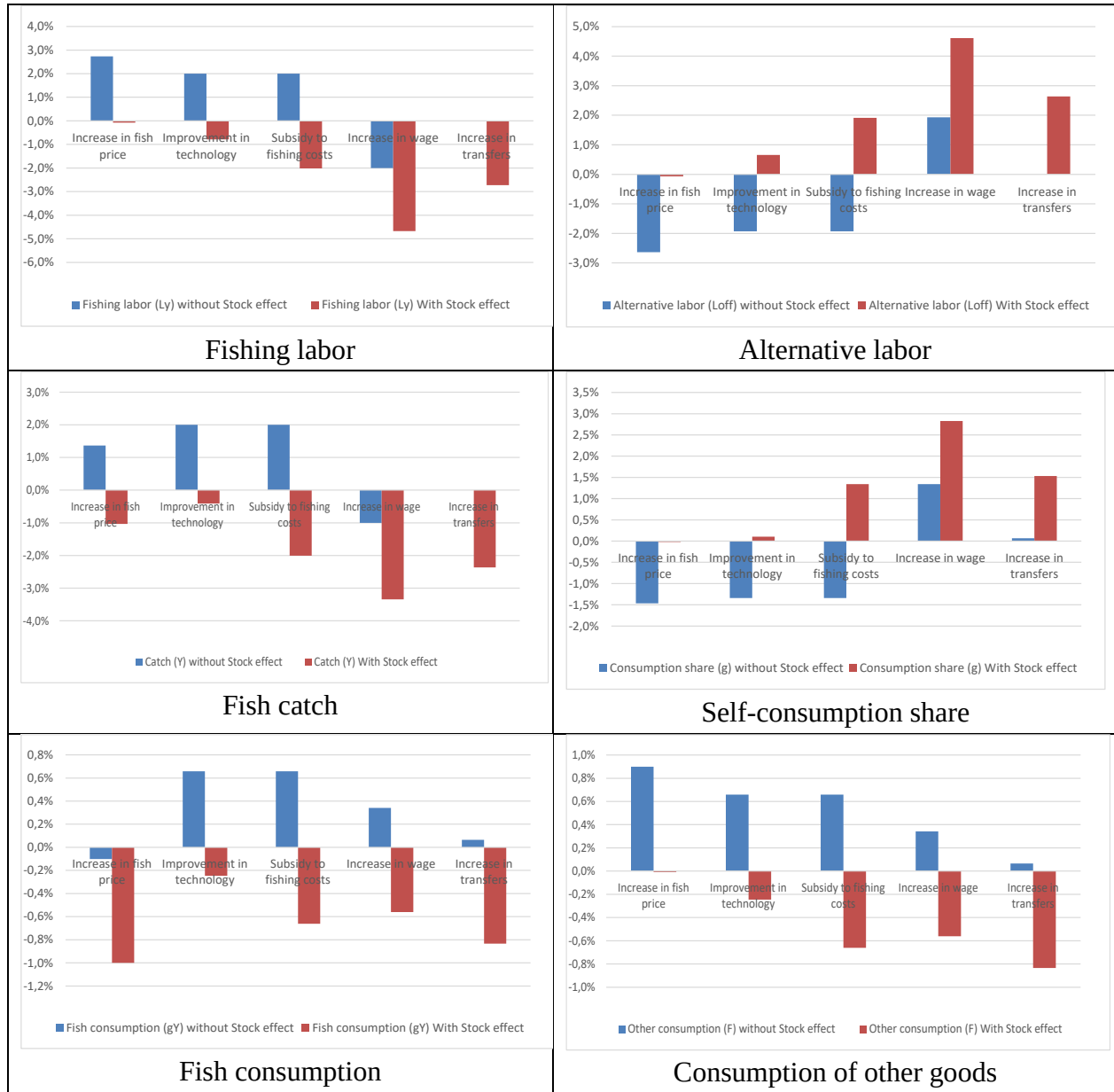
57 When fish stocks decline, the effects of a wage increase are reinforced: fishing activity decreases
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59 further, as does the allocation of household labor to fishing. However, once stock levels decline,
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4 consumption and well-being are negatively affected. We estimate that a compensating variation of
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6 2.66% in wage rate would allow households to recover their original levels of well-being following
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8 a 1% reduction in stock.
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11 When compared with policies that raise fish prices, improve technology, or reduce fishing costs,
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13 the promotion of *alternative activities* is the only intervention that reduces pressure on fish stocks.
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15 In this sense, this policy appears better suited to achieving the dual objective of improving living
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17 conditions while alleviating pressure on natural resources. Examples of such interventions include
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19 a wide range of options, such as the creation of alternative formal jobs in activities related to
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21 ecosystem conservation, restoration, and monitoring, as well as ecotourism. Promoting and
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23 supporting sustainable productive activities may therefore represent a particularly promising policy
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25 alternative.
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31 The panels in Figure 1 compare the effects of the policies with and without accounting for changes
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33 in stock. Stock reductions resulting from overfishing lead to more attenuated—or even reversed—
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35 effects for most variables. Once again, it is important to note that none of the policies restore fish
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37 catch and consumption to their pre-decline levels.
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Figure 1 Comparison of policy effects with (red bars) and without considering stock effects (blue bars) for model variables



6. DISCUSSION

The main purpose of this study was to shed light on the livelihoods of small-scale fishing communities in coastal areas. Specifically, it addresses two questions: (i) How do key economic parameters influence livelihood decisions of fishing households operating in an open-access common-pool resource (CPR) system? and (ii) What policy measures could support both sustainable resource use and local well-being, given that these households depend on a fishery under an open access regime?

The model reveals that the productive variables (total fishing, Y , time spent fishing, L_Y , and time devoted to other activities, L_{Off}) are elastic with respect to changes in all the parameters except fishing costs. In other words, parameter changes affect households' productive decisions more than proportionally. These elastic relationships play an important role in household income generation. By contrast, consumption variables (food acquisition, F , and self-consumption of fish, γY) tend to exhibit elasticities below one, indicating that changes in parameters affect consumption decisions for these goods less than proportionally.

The share of fish allocated to self-consumption, γ , is an elastic variable with respect to most parameters (with the exception of fishing costs, transfers, and the price of other goods). This variable plays a central role in balancing consumption and production decisions, acting as a key adjustment mechanism to support household well-being and sustainability. On the one hand, allocating a portion of the catch to self-consumption enhances household food security. On the other, selling another share of the catch generates income that can be used to purchase food for the household. Kavarazuka & Béné (2010) argue that these decisions are typically complementary. However, the integration of the FHPM suggest that, in some cases, complementarity effects may arise, while in others substitution effects between food sources may dominate. Consequently, the

gamma variable appears to play a key role in bridging household consumption and production decisions.

Our results confirm the presence of the profit effect, as described by Singh et al. (1986b), and its interaction with substitution and income effects in household production models. The coexistence of these three effects implies that, in some cases, the signs of elasticities are not theoretically unambiguous and must therefore be determined empirically.

Overall, the estimated elasticities are consistent with the theoretical predictions in terms of the signs. For elasticities whose signs are theoretically ambiguous, the following empirical findings are particularly noteworthy. In the case of the elasticity of household fish consumption (and the proportion of fish dedicated to self-consumption) with respect to fish price, the empirically observed negative sign indicates that the standard consumer effect dominates the profit effect generated by the increase in income. Another case concerns the elasticity of demand for self-consumed fish and other goods with respect to wages in alternative activities. Although a reduction in catch would typically be expected to lower consumption of both fish and other goods, the positive sign of these elasticities indicates that the profit effect arising from higher income earned in alternative work dominates. As a result, households increase both the consumption of other goods and in the quantity of fish retained for self-consumption. In other words, despite a reduced catch, wage increases lead fishers to keep a larger share of fish for self-consumption.

The model's results can also be used to simulate the effects of different policies. In this case, five policies were analyzed, each affecting key model variables: changes in fish prices, in fishing technology, in the costs associated with fishing activities, in transfers, and in the possibilities of generating income from alternative sources. Among these interventions, the latter—simulated as a change in the external wage rate, w —is the only one that simultaneously delivers gains in fish consumption and household well-being while reducing pressure on natural resources. In contrast,

all other policy options analyzed involve a trade-off between conservation objectives and the economic development of fishing communities.

In a dynamic setting, the presence of a production input that is non-excludable but rival—the fish stock—raises additional considerations. If a policy, for instance, encourages fish extraction, the stock may be affected in the long run, and changes in the fish stock will, in turn, induce adjustments in household decisions. We find that a decline in fish stocks can reduce the effectiveness of policies to the point they become largely irrelevant when compared with original conditions. Therefore, depending on the context of the CPR, households may need to adjust their decisions as the aggregate effects of extraction determine stock size. To maintain the fish stock at its initial levels, fishing must be reduced at a rate that depends on the fleet size and the resource’s growth function. Alternatively, if we treat the proposed policies as compensatory tools, the model suggests that, to restore household well-being after a 1% stock reduction, we would need to increase fish prices by 1.2%, improve technology by 1.38%, raise subsidies by 3.77%, increase wage rates by 2.66%, and raise direct transfers by 13.85%. These figures offer a useful benchmark for estimating the value of the lost fish stock.

These effects are not captured in previous household production models that ignore the non-excludable nature of the fish stock, which is a novel contribution to literature. Notably, higher wages outside the fisheries sector can have a dual positive impact by enhancing both sustainability and fishing households' well-being. Meanwhile, direct transfers may influence consumption without altering fishing-related decisions.

These results have important implications for understanding the relationships among fishing, poverty, and resource conditions, a topic that has been widely discussed in the literature. As noted by Béné (2004), two main interpretations coexist. The first—rooted in Hardin (1968) and Gordon (1954)—views poverty as a consequence of open-access dynamics: overexploitation reduces

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catches, depresses income, and ultimately reinforces poverty (“they are poor because they are
fishers”). The second interpretation highlights factors external to the fishery, emphasizing that
limited access to alternative income opportunities and capital constraints push households into
fishing as a last-resort activity (“they are fishers because they are poor”). Our model is consistent
with both mechanisms: the long-run elasticities reflect the income dissipation typical of open
access regimes, while the short-run elasticities reveal that restricted off-fishing opportunities keep
households dependent on fishing for income and food security.

For a community like Barú—with no access to land, limited access to credit, and few alternative
income-generating activities—strategies should be developed to address these constraints as part
of broader conservation and development planning. As shown in this study, a significant proportion
of the community's fishing households live just above the poverty line and therefore do not qualify
for direct state support, such as cash transfers. Imposing restrictions on fishing would increase
poverty and vulnerability, posing a threat to food security. This, in turn, would demand targeted
attention from the state to expand social protection programs and help mitigate the adverse effects
on income and food security. Unfortunately, these conditions are not unique to Barú and apply to
many other communities across the developing world.

In this context, fishing plays a dual role as both a source of income and a safety net for vulnerable
households. Restricting access without providing viable alternatives can increase poverty and food
insecurity, as it removes a critical subsistence buffer. Conversely, allowing further resource
depletion would eventually erode this safety net, as declining stocks reduce catches and household
food intake, replicating the impoverishment mechanism described in the open-access literature.
Our results suggest that neither strict restrictions nor unchecked depletion are welfare-enhancing;
rather, effective policies must simultaneously address resource sustainability and the underlying
economic constraints that limit livelihood diversification in small-scale fishing communities.

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